

SMART INDIA HACKATHON 2026

Sahayak CoopServe

A Cooperative-Owned Digital Marketplace for India's Gig Workforce

Detailed Project Report

Field	Detail
Problem Statement ID	26089
Problem Statement Title	Cooperative Gig Services Platform for Household & Community Services
Theme	Agriculture, FoodTech & Rural Development
PS Category	Software
Team Name	Tech Team
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Source Code	github.com/aditya1shahi/Gig_Workers_SIH_PS89
Live Prototype	gig-workers-sih-ps-89.vercel.app

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1. Executive Summary

Sahayak CoopServe is a cooperative-owned digital marketplace that connects verified skilled workers, organised through registered labour associations, with households and institutions that need their services. The platform is designed so that the surplus generated by the marketplace returns to the worker-members who create it, rather than to external shareholders.

India's gig workforce is large and growing. NITI Aayog estimates 7.7 million gig workers in 2020–21, projected to reach 23.5 million by 2029–30 [Ref. 1]. Yet the hiring of local service workers remains largely informal: there is no dependable verification layer, no wage floor, no portable work history, and no welfare cover. Where digital platforms do intermediate this work, they typically retain a commission of 15–30% of the job value.

Our response is a cooperative-first platform. Registration is anchored to labour associations, so verification has an accountable institutional owner. Pricing rules are set by worker-members rather than by a platform operator. Reputation is designed to belong to the worker and travel with them. Welfare provisions — insurance, emergency credit, training — are treated as first-class platform features rather than optional extras.

A working prototype is deployed and publicly accessible at gig-workers-sih-ps-89.vercel.app under the name GigWorker Connect. It supports association registration, worker registration with government ID capture, service search across six trades and seven cities, a live map of worker locations built on Leaflet and OpenStreetMap, worker profiles with rates and ratings, and capture of job details for an assignment. Two significant components are not yet built: persistent database storage (the deployed demo runs on hardcoded data) and payment gateway integration. This report states that boundary explicitly wherever prototype capability is described.

Scope of evidence in this report

Every quantitative claim in this document is either cited to a published source or is labelled as a projection with its derivation shown. No usage figures, performance benchmarks, revenue figures, partnerships or pilot results are claimed, because none exist yet. Fields that the team has not yet supplied are marked [INFORMATION REQUIRED] rather than filled with plausible-looking values.

2. Problem Statement and Context

2.1 The scale of the affected population

NITI Aayog's report India's Booming Gig and Platform Economy (June 2022) estimates the gig workforce at 68 lakh (6.8 million) in 2019–20 and 77 lakh (7.7 million) in 2020–21, forming approximately 2.6% of the non-agricultural workforce. The same study projects growth to 2.35 crore (23.5 million) workers by 2029–30, or about 6.7% of the non-agricultural workforce [Ref. 1].

The report is explicit that this is an indirect estimate derived from proximate worker characteristics, and that no official enumeration of gig workers currently exists in India. We reproduce that caveat rather than presenting the figure as a precise census.

2.2 The official problem statement

Problem Statement 26089 is stated as follows.

Background: Labour Cooperative Federations and Labour Cooperative Societies possess a large pool of skilled workers such as electricians, plumbers, carpenters, painters, domestic helpers, caregivers, drivers, gardeners, cleaners and technicians. However, they lack a structured digital platform to connect these workers with households and institutions requiring such services. Private platforms currently dominate this market, while cooperative workers often remain underutilised despite having skills and local presence.

Problem: To develop a cooperative-owned digital service marketplace platform that enables Labour Cooperative Federations and Labour Cooperative Societies to provide verified household and community services while ensuring fair wages, worker welfare and consumer trust.

The expected solution defines eleven features and five technology components. Section 6.3 maps each of them against what the prototype currently delivers.

2.3 Who experiences the problem

Stakeholder	Experience of the current system
Skilled workers	Income depends on physical labour chowks, word of mouth and intermediaries. No portable proof of skill or work history. No insurance, wage floor or grievance route.
Households and institutions	No dependable way to establish whether a worker is who they claim to be, is competent, or is accountable after the job. Hiring rests on personal referral.
Labour cooperatives and associations	Hold real membership and verification knowledge but have no digital channel through which to convert it into work for their members.
Government	The Social Security Code, 2020 recognises gig and platform workers, but delivery of entitlements depends on workers being identifiable and reachable, which informal hiring prevents.

2.4 Limitations of the current alternatives

- Commission-led platforms: private aggregators intermediate the work but retain 15–30% of job value, which is deducted from worker earnings.
- Directory listings and classifieds: cheap to operate but provide no verification, no booking state and no accountability once contact details are exchanged.
- Physical labour chowks: no scheduling, no record, and highly exposed to weather, seasonality and the discretion of contractors.
- Government portals: authoritative for entitlements, but not designed as day-to-day demand-matching marketplaces for household service work.

2.5 Consequence if the problem is not addressed

Three effects compound. Workers remain invisible to formal systems, so credit, insurance and welfare entitlements stay out of reach even where a statutory basis exists. Consumers continue to carry the full risk of unverified hiring. And the value created by local service work continues to leak out of local economies through commission, rather than being reinvested in the workforce that generates it.

3. Proposed Solution

3.1 Concept

Sahayak CoopServe is a multilingual, cooperative-owned marketplace with three participant types: labour associations, workers who belong to those associations, and customers who need a service. Associations register first and act as the verification anchor; workers register against an association; customers search by service and city, review available workers, and issue an assignment with job details.

The governing idea is that ownership determines behaviour. A platform owned by worker-members sets its pricing, commission and dispute rules through those members, which structurally removes the incentive to increase take-rate at the expense of worker earnings.

3.2 How the design addresses each element of the problem

Problem	Design response
Informal, unverifiable hiring	Association-anchored registration. A worker profile is tied to a registered association with a registration number, contact person and address, giving

Problem	Design response
	verification an accountable institutional owner.
15–30% commission extraction	Cooperative ownership. Surplus is retained by the member body and directed to welfare, training and cooperative development instead of external shareholders.
Reputation locked to a platform	Worker-owned reputation. Ratings and job history are modelled as attributes of the worker record, designed to be portable across cities and associations.
No welfare or safety net	An embedded welfare layer — insurance, emergency credit and training — planned as platform features funded from retained surplus.
Digital access barriers	Low-bandwidth, vernacular and voice-first interfaces planned for the production build, so participation does not require a high-end device or English literacy.
Safety concerns for women workers	Planned SOS support and female-only request routing for women caregivers and service providers.

3.3 Innovation and differentiation

1. Cooperative-first governance. Most comparable platforms are investor-owned. Rules and pricing here are set by the worker-members themselves.
2. Association as verification primitive. Rather than attempting to verify millions of individuals directly, the platform verifies associations and inherits trust through membership — a far more tractable trust problem at national scale.
3. Portable worker reputation. Work history is designed as an asset the worker carries, not as data the platform retains as a lock-in mechanism.
4. Welfare as core function. Insurance, emergency credit and skilling are treated as platform responsibilities funded by retained surplus, not as marketing add-ons.
5. Built for Bharat. Vernacular, low-bandwidth and offline-tolerant design is a requirement of the production build rather than a later localisation exercise.

4. System Design

4.1 End-to-end process flow

The following sequence is implemented in the deployed prototype, with the exception of step 6, which is not yet integrated.

#	Stage	What happens
1	Worker / association registration	An association registers with its name, registration number, contact person, phone, email, address, city and state. Workers then register against an association with their trade, experience, service area, government ID number and hourly rate.
2	Service search	The customer selects a service category and a location.
3	Worker discovery	The system returns workers matching the selected service and location.
4	Worker selection	The customer reviews the available workers and selects one.
5	Job details capture	The customer supplies their name, phone number, service address and a job description.
6	Booking payment	Not integrated in the current prototype. Planned as UPI and gateway-based payment with a cooperative wallet, alongside a cash-with-receipt path for cash-dominant contexts.
7	Assignment confirmed	The worker is assigned to the job and the customer receives confirmation.

4.2 Architecture

The prototype is a three-tier web application. The presentation tier is served as static HTML and CSS; the application tier is a Node.js service using the Express framework, which holds routing and business logic and issues JWT-based sessions; the data tier is MongoDB, for which the schema has been designed but which is not yet wired to the running deployment.

Tier	Component	Responsibility
Presentation	HTML, CSS	Registration forms, service search, worker listing, assignment dialog. Rendered in the browser with no framework dependency.
Application	Node.js with Express	Request routing, validation, search and matching logic, session issue and verification.
Session / identity	JWT (JSON Web Token)	Stateless authentication of association, worker and customer sessions.
Mapping	Leaflet + OpenStreetMap	Renders a live map of worker locations for the selected city alongside the results list.
Data	MongoDB	Intended store for association, worker, service and booking records. Schema designed; not yet connected — the deployed demo serves hardcoded data.

4.3 Principal entities

The following entities are derived from the fields captured by the deployed prototype's forms.

Entity	Attributes captured
Association	Association name, registration number, contact person, phone number, email address, address, city, state.
Worker	Full name, phone number, email address, service category, years of experience, associated association, service area, government ID number, hourly rate.
Booking / assignment	Selected worker, customer name, customer phone number, service address, job description.
Service catalogue	Six trades: electrician, plumber, carpenter, painter, mechanic, cleaner.
Service area	Seven cities: Delhi, Mumbai, Bangalore, Noida, Chennai, Pune, Hyderabad.

4.4 Modules

- Registration module — association onboarding and worker onboarding, including government ID capture and association linkage.
- Discovery module — service category and location filtering, returning the matching worker set.
- Assignment module — worker selection, job detail capture and assignment confirmation.
- Identity module — JWT issue and verification for authenticated sessions.
- Payments module — *designed, not yet implemented. Intended to cover UPI, gateway payments, a cooperative wallet and a cash-with-receipt path.*
- Welfare module — *designed, not yet implemented. Intended to cover insurance enrolment, emergency credit and training recommendations.*

5. Technology Stack

The prototype stack was chosen for low operational cost and wide availability of skills, both of which matter for a cooperative that must eventually own and maintain the system itself.

Layer	Technology	Rationale
Frontend	HTML, CSS	No build toolchain and minimal payload, which suits low-bandwidth users and keeps the prototype auditable.
Backend	Node.js (Express)	Single-language stack with the frontend, large talent pool, minimal framework overhead for a REST-style API.
Authentication	JWT (JSON Web Token)	Stateless sessions, straightforward to scale horizontally without shared session storage.
Database	MongoDB	Document model suits evolving worker and association records during early iteration. Schema designed; integration pending.
Maps	Leaflet + OpenStreetMap	Open, royalty-free mapping with no per-request billing, unlike commercial map APIs.
Hosting	Vercel	Zero-cost deployment for the prototype with a public URL for evaluation.

Every component in the stack is free or open-source. There is no licensing cost in the prototype, and no element of the design depends on an unsolved research problem — the remaining work is integration rather than invention.

5.1 Technologies identified for the production build

The following are selected for the full implementation but are not part of the current prototype. They are listed here as roadmap, not as delivered capability.

Area	Intended technology
Mobile application	React Native or Flutter
Geo-spatial matching	Mapbox / Leaflet with spatial indexing
Payments	Razorpay or UPI rails, plus a cooperative wallet
Notifications	Firebase Cloud Messaging and SMS
Matching intelligence	Demand forecasting and skill/geo matching models

6. Prototype Status and Evidence

A prototype is deployed and publicly reachable. This section states precisely what it does and does not do, so that the claim can be checked directly rather than taken on trust.

Item	Detail
Deployed application	GigWorker Connect — gig-workers-sih-ps-89-aditya1shahis-projects.vercel.app
Source repository	github.com/aditya1shahi/Gig_Workers_SIH_PS89
Service categories live	Electrician, plumber, carpenter, painter, mechanic, cleaner (6)
Service areas live	Delhi, Mumbai, Bangalore, Noida, Chennai, Pune, Hyderabad (7)
Map layer	Leaflet with OpenStreetMap tiles, showing worker locations for the selected city
Screenshots	Four, reproduced in section 6.4

6.1 Implemented

- Association registration form and submission flow.
- Worker registration form, including trade, experience, association linkage, service area, government ID and hourly rate.
- Service and location search returning a worker listing.
- Worker selection and job detail capture (customer name, phone, service address, job description).
- Assignment confirmation in the user interface.
- Live map of worker locations for the selected city (Leaflet + OpenStreetMap).
- Worker profile cards showing trade, rating, association, hourly rate and availability.
- JWT-based session handling.

6.2 Not yet implemented

- **Persistent storage.** The deployed demo runs on hardcoded data; MongoDB integration is designed but not connected.
- **Payment gateway integration** for the booking amount.
- **Live dashboard metrics.** The counters on the landing page are placeholders and are not populated.
- **AI-based geo and skill matching;** the current search is attribute filtering.
- **Mobile application and vernacular interface.**
- **Welfare, insurance and emergency credit features.**

6.3 Coverage of the expected solution features

Problem Statement 26089 lists eleven expected features. The table below states, feature by feature, what exists in the prototype today and what remains. Nothing is marked delivered unless it can be exercised in the deployed application.

Expected feature	Status	Evidence / plan
Service provider registration and verification	Partial	Association and worker registration are live, with government ID capture and association linkage. Automated verification against ID databases is not built.
Worker skill profiling and certification	Partial	Trade, years of experience, service area and hourly

Expected feature	Status	Evidence / plan
		rate are captured. Formal certification records are not yet modelled.
Customer booking and scheduling system	Partial	Booking through worker assignment with job details is live. Calendar scheduling and slot management are not built.
Geo-location based service matching	Partial	City-based filtering with a live Leaflet / OpenStreetMap view of worker locations is live. Distance-optimised matching is not built.
Digital payments and invoicing	Planned	Not integrated. UPI and gateway payments with a cooperative wallet and a cash-with-receipt path are designed.
Rating and feedback mechanism	Partial	Ratings and job counts are displayed on worker cards. Customer-submitted feedback after a job is not built.
Worker welfare and insurance integration	Planned	Designed as a platform layer funded from retained surplus. Not implemented.
Emergency and on-demand service booking	Planned	Availability status is shown per worker; priority and emergency routing are not implemented.
Cooperative federation administration dashboard	Planned	Landing-page counters exist but are not driven by live data. A federation-level administration view is not built.
Multilingual mobile application	Planned	Current build is a responsive web application in English. React Native or Flutter with vernacular support is the intended path.
AI-based demand forecasting and workforce allocation	Planned	Not implemented. Identified as a medium-term objective once real booking data exists.

Of the eleven expected features, six are partially delivered in a running prototype and five remain at design stage. We consider an honest split of this kind more useful to an evaluator than a uniform claim of completion.

6.4 Prototype screenshots

The following are captured from the deployed application.

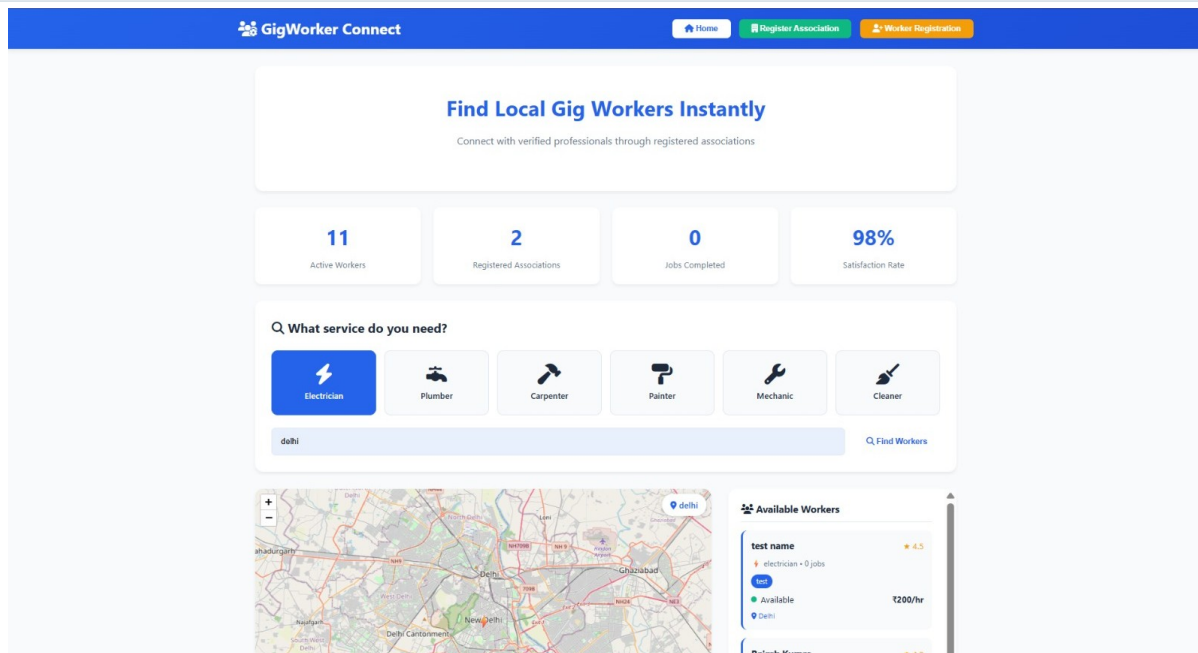


Figure 1 — Landing page: service categories, city search and platform counters.

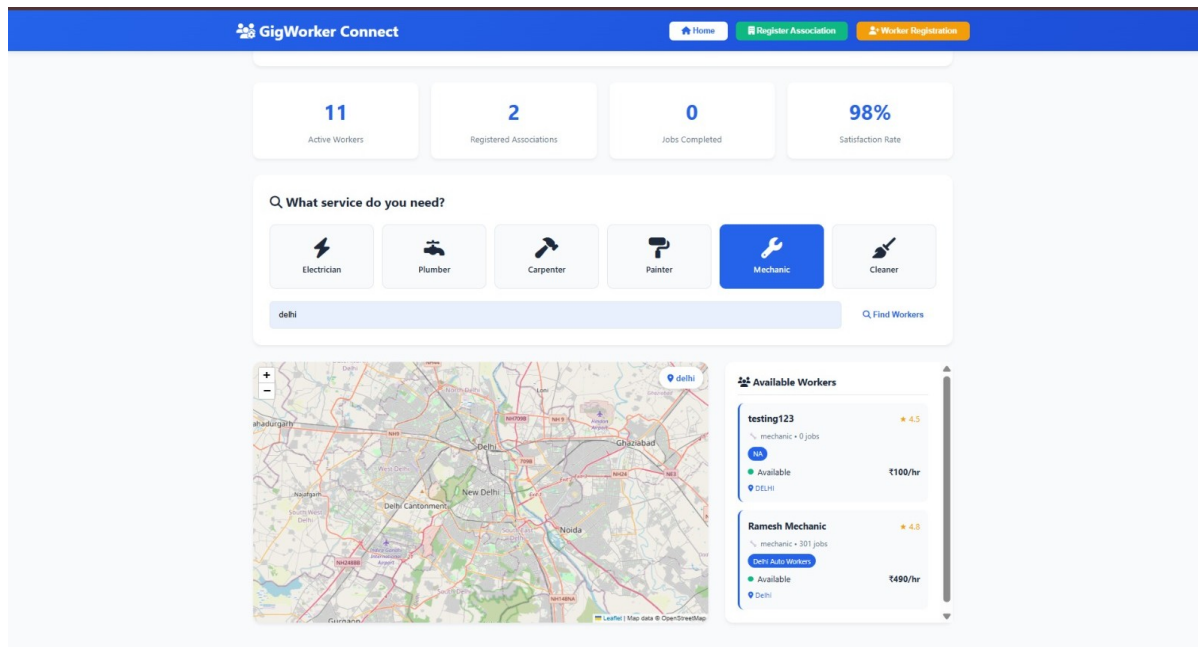


Figure 2 — Search results: available workers with trade, rating, association, rate and availability, alongside a live Leaflet / OpenStreetMap view.

GigWorker Connect Home Register Association Worker Registration

Register as a Worker
Data will be saved to MongoDB

Full Name * Phone Number *

Email Address *

Service Category * Years of Experience *

Select service

Associated Association *

Association name

Service Area *

Select city

Government ID Number *

Aadhaar / PAN / Voter ID / DL number

Hourly Rate (₹) *

Register as Worker

Figure 3 — Worker registration: trade, experience, association linkage, service area, government ID and hourly rate.

GigWorker Connect Home Register Association Worker Registration

Register Your Association
Data will be saved to MongoDB

Association Name * Registration Number *

Contact Person * Phone Number *

Email Address *

Address *

City * State *

Register Association

Figure 4 — Association registration: the verification anchor for every worker profile.

Action required before submission

The deployed application currently displays the text “Data will be saved to MongoDB” on its registration forms and “Saved to MongoDB” on success. Because storage is not yet connected, this wording overstates current capability and should either be changed to indicate demo mode, or the MongoDB integration should be completed before evaluation.

7. Feasibility and Viability

7.1 Technical feasibility

The end-to-end journey from registration to assignment already runs in a deployed build, which establishes that the core interaction model works. The remaining technical work — connecting MongoDB, integrating a payment gateway, and adding matching intelligence — uses established, well-documented components. No part of the roadmap requires a novel algorithm or unproven infrastructure.

7.2 Economic feasibility

The prototype currently runs at no licensing or hosting cost: the entire stack is open-source and the deployment tier is free. The economic argument for the model rests on the commission differential. Where a private platform retains 15–30% of job value, a cooperative structure retains that margin within the member body.

Cost line	Status
Software licensing	Nil — open-source stack
Prototype hosting	Nil — free deployment tier
Production infrastructure estimate	[INFORMATION REQUIRED]
Payment gateway transaction fees	[INFORMATION REQUIRED]
Verification and onboarding cost per worker	[INFORMATION REQUIRED]
Insurance and welfare fund contribution model	[INFORMATION REQUIRED]

These figures are left open deliberately. Producing them requires quotations and a pilot cohort size, and inventing them would undermine the rest of the analysis.

7.3 Operational feasibility

- Assisted onboarding through on-ground “Coop Champions” drawn from the associations themselves, addressing the digital literacy gap directly rather than assuming it away.
- Hybrid payment handling — UPI, cash-with-receipt and a cooperative wallet — so that a cash-preferring worker is not excluded.
- Tiered verification: basic identity first so that onboarding is not blocked, with skill certification layered on subsequently.
- Association-led maintenance: because the stack is common and open-source, a cooperative can eventually hire for it without depending on a single vendor.

8. Risk Register

Risk	Impact	Mitigation
Worker digital literacy	High	On-ground Coop Champions provide assisted onboarding; interface designed for low literacy and low bandwidth.
Trust and adoption on both sides of the marketplace	Medium	Trust campaigns built around cooperative ownership and member benefit; association-anchored verification as the visible guarantee.
Payment reconciliation in a cash-dominant sector	Medium	Hybrid model combining UPI, cash-with-receipt and a cooperative wallet, with reconciliation at the association level.
Scaling the verification process	High	Tiered verification — basic identity first, skill certificates later — with associations performing first-line verification.
Regulatory compliance (DPDP Act, labour law)	Medium	Compliance-by-design: data minimisation, purpose limitation and consent capture from the first production release.
Prototype overstating capability	Medium	Explicit labelling of unimplemented features in all materials; correction of in-app copy that implies persistence.
Dependence on association participation	Medium	Pilot design begins with associations that already have active membership, so the model is tested where the precondition holds.

9. Impact and Benefits

9.1 Projected effect on worker earnings

The clearest quantifiable benefit is the removal of platform commission. The table below is a projection derived arithmetically from the commission range that private platforms charge. It is not a measured result and is not based on observed transactions on our platform.

Scenario	Commission	Worker take-home per ₹100 billed
Private platform, upper end of range	30%	₹70
Private platform, lower end of range	15%	₹85
Sahayak CoopServe (cooperative model)	0%	₹100

Derivation: worker take-home = ₹100 less the stated commission. The 15–30% range is the commission band attributed to private platforms in the project research. Actual cooperative take-home will be reduced by operating costs and any welfare-fund contribution agreed by the members, which are not yet quantified.

9.2 Qualitative comparison

Today	With Sahayak CoopServe
Informal, word-of-mouth hiring	Verified, app-based booking
15–30% commission extracted from each job	Cooperative-owned; surplus retained by members
No insurance or welfare cover	Insurance, credit and training designed into the platform
Reputation lost when a worker changes platform	Portable reputation held by the worker
Pricing set unilaterally by the platform	Pricing set by worker-members

9.3 Benefits by stakeholder

Stakeholder	Benefit
Workers	Higher net earnings through commission removal; formal work history; access to welfare provisions; a portable professional identity.
Customers	Verified, association-accountable workers with a traceable booking record.
Cooperatives	A digital channel that converts existing membership and verification capability into work for members, with surplus retained locally.
Government	A route to making gig workers identifiable and reachable, supporting delivery of entitlements recognised under the Social Security Code, 2020.
Women workers	Planned SOS support and female-only request routing, addressing a specific and documented barrier to participation.
Environment	Reduced travel through location-aware matching, and paperless invoicing and records.

Environmental and safety benefits above are design intentions of the production build, not measured outcomes of the prototype.

10. Regulatory and Compliance Position

Two instruments are directly relevant to this platform.

The Social Security Code, 2020 introduces statutory recognition of gig workers and platform workers, defining a gig worker as a person who earns from a work arrangement outside a traditional employer-employee relationship. This recognition is the legal foundation on which welfare delivery to platform workers can be built, and it aligns the platform's welfare layer with an existing policy direction rather than requiring new legislation.

The Digital Personal Data Protection Act, 2023 governs the handling of the personal data this platform necessarily collects — including government identification numbers, contact details and service addresses. The design commitments that follow from it are data minimisation, explicit purpose limitation, consent capture at the point of collection, and retention limits. Because the platform captures a government ID number during worker registration, this is a material compliance obligation and not a formality.

Compliance work outstanding

A formal data protection assessment, a published privacy notice, a consent record design and a retention schedule are required before any production deployment handling real personal data. None of these exist yet.

11. Scalability Path

Stage	Scope	Prerequisites
Prototype	Current: deployed demo covering six trades and seven cities	Complete database integration and payments
Pilot	One to two cities with participating labour associations	Association partnerships; onboarding cohort; cost model
Multi-city	Extension to neighbouring cities using the pilot playbook	Repeatable onboarding process; support capacity
State-level	Formal engagement with state labour departments	Compliance package; integration with state welfare schemes
National framework	A reference model for cooperative platform work	Demonstrated outcomes at state level; policy engagement

12. Future Scope

12.1 Immediate (next development cycle)

- Connect MongoDB and remove hardcoded data.
- Integrate the payment gateway and cooperative wallet.
- Populate live dashboard metrics from real records.
- Correct in-app copy that implies persistence not yet present.

12.2 Medium term

- Mobile application with vernacular and voice-assisted interfaces.
- Geo-aware matching to reduce worker travel and improve response time.
- Ratings, reviews and the portable reputation record.
- Insurance enrolment and emergency credit integration.

12.3 Long term

- Demand forecasting to help associations plan workforce allocation.
- Federation across cooperatives so reputation and membership travel between cities.

- A reference implementation that other cooperatives can adopt, positioning cooperative technology as a viable alternative to investor-owned platforms.

13. References

All links below were checked and resolve to the stated source.

Policy, law and government research

#	Source	Link
1	NITI Aayog (2022). India's Booming Gig and Platform Economy: Perspectives and Recommendations on the Future of Work.	niti.gov.in/sites/default/files/2022-06/25th_June_Final_Report_27062022.pdf
2	Digital Personal Data Protection Act, 2023 — statutory text.	indiacode.nic.in/indiacode/handle/123456789/22037
3	Ministry of Electronics and Information Technology — DPDP Act, 2023.	meity.gov.in/content/digital-personal-data-protection-act-2023
4	Digital Labour Chowk — private blue-collar hiring platform (India). Benchmark, not a government portal.	web.dlc.org.in

Research on AI and labour demand

#	Source	Link
5	Hampole, Papanikolaou, Schmidt & Seegmiller (2025). Artificial Intelligence and the Labor Market. NBER Working Paper 33509.	nber.org/papers/w33509
6	Acemoglu & Restrepo (2018). Artificial Intelligence, Automation and Work. NBER Working Paper 24196.	nber.org/papers/w24196
7	Liu, Wang & Yu (2025). Labor Demand in the Age of Generative AI. World Bank Policy Research Working Paper 11263.	openknowledge.worldbank.org/entities/publication/b5d5c33c-9419-4dc7-93fb-ec9d6b997c4b
8	World Bank (2025). Digital Progress and Trends Report 2025.	documents1.worldbank.org/curated/en/099112525160536089
9	DSIT / DCMS (2026). AI Skills for Life and Work: summary report. Published 28 January 2026.	gov.uk/government/publications/ai-skills-for-life-and-work-summary-report

Technology documentation

#	Source	Link
10	Node.js	nodejs.org
11	Express (Node.js web framework)	expressjs.com
12	MongoDB documentation	mongodb.com/docs
13	JSON Web Token	jwt.io
14	React Native (planned mobile build)	reactnative.dev
15	Flutter (planned mobile build)	flutter.dev

#	Source	Link
16	Razorpay documentation (planned payments)	razorpay.com/docs
17	Mapbox (planned geo-spatial)	mapbox.com
18	Firebase Cloud Messaging (planned notifications)	firebase.google.com/docs/cloud-messaging

Appendix A — Outstanding Items

The following must be resolved before this report and the accompanying presentation are submitted.

Item	Owner	Status
Problem Statement ID, title, theme and category	Team	Confirmed
Institution, team leader and member names	Team	Confirmed
Prototype screenshots as build evidence	Team	Supplied
Live deployment publicly reachable	Team	Verified
Team ID as registered on the SIH portal	Team	To add on the portal form
Correct in-app copy implying MongoDB persistence	Team	Open
MongoDB integration to replace hardcoded data	Team	In progress
Payment gateway integration	Team	Planned
Production infrastructure and API cost estimates	Team	Open
Demo video walkthrough	Team	Optional

Appendix B — Glossary

Term	Meaning
Gig worker	A person who earns from a work arrangement outside a traditional employer-employee relationship, as defined in the Social Security Code, 2020.
Platform worker	A gig worker whose work is accessed through an online platform.
Labour cooperative / association	A registered member body of workers, which in this design acts as the verification anchor for its members.
Commission / take-rate	The share of job value retained by a platform operator; 15–30% for private platforms in the project research.
Portable reputation	Ratings and work history modelled as belonging to the worker rather than to a platform.
JWT	JSON Web Token — a signed token format used to maintain authenticated sessions without server-side session storage.
DPDP Act	The Digital Personal Data Protection Act, 2023, governing the processing of personal data in India.
Coop Champion	An on-ground assistant drawn from a partner association who helps workers onboard to the platform.